

Catherine Zhao

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EDUCATION

Massachusetts Institute of Technology (MIT)

Expected May 2029

- **GPA:** 5.0/5.0; Candidate for a Bachelor of Science in *Computer Science and Engineering*
- **Relevant Coursework:** Linear Algebra & Optimization; Algorithms; Machine Learning; C & Low-Level Assembly; Shape Analysis (grad); Deep Learning (grad); Computation Structures; Probability and Random Variables

TECHNICAL SKILLS

Languages: Python, C, Java, JavaScript, TypeScript, HTML/CSS, Julia, Assembly, SQL

Tools/Frameworks: Git/GitHub, React, Flask, VS Code, PlatformIO, Vite, ITK-Snap, Blender, Perforce

Systems & HPC: Unix, Bash, HPC cluster workflows, SSH/SSHFS, SLURM

WORK EXPERIENCE & RESEARCH

MIT EECS - 6.390 Intro to Machine Learning Lab Assistant

Sept. 2026 - Present

- Performed lab checkoffs and held office hours to strengthen students' understanding of machine learning concepts

InterSystems, Boston, MA - Application Services Intern

May 2026 - Aug. 2026

- Developed custom Jira project and CCR type to manage hotfix tracking across TrakCare editions
- Built integration logic between Jira, CCR, and Perforce, enabling synchronized workflow transitions, automated file retrieval and change control, and streamlined code deployment and migration across branches.

Medical Vision Group, MIT CSAIL - Undergraduate Researcher

Jan. 2026-May 2026

- Developed automated pipeline for reconstruction of 3D placenta shape from ~36k MRI scan volumes via marching cubes, PCA alignment, convex hull fitting, and SDF analysis to support maternal-health research.

PROJECTS

ScribbleHero, MIT Reality Hackathon (Team of 4)

Jan. 2026

- Developed *ScribbleHero*, an AR drawing puzzle game built for Snap Spectacles using Snap Lens Studio and Google Gemini. Won honorable mention in the Game Prototyping Track.
- Implemented event-driven systems in Typescript to manage user interaction and physics-based animations.

Tactile 3D Viewer, Assistive Technology Club (Team of 5)

Oct. 2025 - Jan.2026

- Built a full-stack web tool allowing legally blind users to explore 3D CAD models via 2D tactile slices.
- Used Trimesh for ray-mesh intersection and slicing, with caching for low-latency rendering.

LEADERSHIP & ENGAGEMENT

MIT Assistive Technology Club - Project Manager

Feb. 2026-Present

- Organized Git & software principles training workshop, helped plan study breaks and club-wide events, designed promotional content for club outreach, represented club at the Boston Museum of Science Disability Pride Event.

MIT Asian Dance Team - Choreographer, Cover Director (prev), Promo Maker (prev)

Mar. 2026-Present

- Co-led a team of 8 dancers to perform and film a dance cover for the MIT ADT YouTube channel; taught choreography, oversaw editing, designed promotional content.

MIT Admissions - Tour Guide

Feb. 2026-Present

- Explained research, academic, and extracurricular opportunities at MIT to groups of ~30 campus visitors.